The Lumen Moth Keeper's Handbook

A handbook for keepers of lumen moths, invented glowing moths kept in glass houses for the light they give.

1. Introduction to Lumen Moth Biology

Learning objectives

LO-1.1.1 Identify the primary anatomical structures of the Lumen Moth. (K1)

LO-1.1.2 Explain the function of the thorax in light generation. (K2)

LO-1.1.3 Distinguish between male and female antenna structures. (K2)

LO-1.2.1 List the three stages of the Lumen Moth life cycle. (K1)

LO-1.2.2 Describe the transition process from Nymph to Imago. (K2)

LO-1.2.3 Calculate the expected incubation time of a Dormant Egg at 40 Vels. (K3)

LO-1.3.1 Define photogenesis in the context of the Lumen Moth. (K1)

LO-1.3.2 Explain the chemical reaction between lumium and atmospheric oxygen. (K2)

LO-1.3.3 Diagnose the cause of irregular pulsing in an adult moth's light. (K3)

LO-1.4.1 State the duration of the standard breeding cycle. (K1)

LO-1.4.2 Summarize the effects of the Frost Moon on egg production. (K2)

LO-1.4.3 Adjust breeding conditions to counteract a delayed seasonal shift. (K3)

LO-1.5.1 Recall the four standard flight patterns of the swarm. (K1)

LO-1.5.2 Interpret the meaning of the "spiral descent" behavior. (K2)

LO-1.5.3 Formulate a response plan for swarm aggregation in the upper corners. (K3)

Welcome to the foundational text of the Keepers Guild, designed to introduce apprentices to the biological marvel that is Lepidoptera luminis. This chapter provides a comprehensive overview of the moth's anatomy, life cycle, photogenic capabilities, and behavioral instincts. Mastery of these biological principles is the first step toward successfully managing a thriving, luminous swarm within a controlled glass house environment.

1.1. The Anatomy of the Lumen Moth

The study of the Lumen Moth begins with a thorough understanding of its unique physical structure. Native exclusively to the harsh, shadowed valleys of the Alabaster Mountains, the species has evolved a highly specialized physiology to survive and communicate in perpetual twilight. Every Keeper must intimately understand these anatomical structures to ensure the health and productivity of their swarm. The primary anatomical structures of the insect include the head, the specialized thorax, the abdomen, the delicate wings, and the sensory antennae. Each component plays a vital role in the creature's survival and its ability to generate its signature glow. The head houses the compound eyes and the proboscis, which is uniquely adapted to siphon the specialized synthetic nectar brewed in the glass houses. Below the head lies the abdomen, a segmented region responsible for digestion, respiration through tiny spiracles, and reproduction. However, the most critical structure for a Keeper to understand is the thorax, which is heavily modified compared to other moth species.

The thorax serves as the biological combustion chamber for the insect's light generation. It houses the

lumium glands, which are surrounded by a dense network of tracheal tubes and reinforced muscular tissues. This central region is responsible for drawing in atmospheric oxygen and mixing it with lumium, a specialized organic compound synthesized from the moth's diet. The thorax is encased in a translucent chitinous shell that protects the internal organs while allowing the maximum amount of light to escape. The muscular contractions within the thorax not only power the wings but also regulate the flow of oxygen into the lumium glands, directly controlling the intensity and rhythm of the light. A healthy adult thorax will appear slightly swollen and emit a soft, continuous baseline glow even when the moth is resting. Keepers must regularly inspect the thorax of fallen moths to ensure the lumium glands are not hardening, a common sign of dietary imbalance.

Differentiating between the sexes is another fundamental skill for any Keeper, and this is done most accurately by examining the antenna structures. Male and female moths possess distinctly different antennae, adapted to their specific roles within the breeding cycle. Male antenna structures are broad, feathery, and comb-like, featuring wide-spanning sensory filaments. These intricate filaments are highly sensitive and are designed to detect the faintest traces of female mating pheromones, as well as the airborne spores of the moon-root plant, across vast distances in the wild. In contrast, female antenna structures are slender, smooth, and needle-like. Instead of tracking pheromones, the female's antennae are highly attuned environmental sensors geared towards measuring ambient humidity and temperature. This precision allows the female to locate the exact microclimate required for laying her eggs. Misidentifying the sex of a moth can lead to skewed breeding populations, so apprentices must spend hours in the observation tents learning to spot the feathery male antennae and the needle-like female antennae at a glance. Together, these anatomical marvels form a creature perfectly adapted to its environment and indispensable to the Guild of Oakhaven.

1.2. The Life Cycle: From Egg to Imago

1.2.1. The Dormant Egg

The life cycle of the Lumen Moth progresses through three distinct stages of development, beginning with the Dormant Egg, transitioning into the Crawling Nymph, and culminating in the Glowing Imago. Understanding each phase is crucial for maintaining a stable population within the glass house. The first stage, the Dormant Egg, appears as a tiny, iridescent sphere deposited on the bark of Iron-Oak branches. In their natural habitat within the Alabaster Mountains, these eggs endure freezing temperatures by entering a state of complete metabolic suspension. In the controlled environment of a glass house, Keepers must carefully manage the climate to trigger healthy development. The expected incubation time of a Dormant Egg at exactly forty Vels is precisely fourteen days. To calculate the incubation schedule, a Keeper simply marks the calendar fourteen days from the moment of oviposition, provided the temperature remains steady at forty Vels. If the temperature drops, the calculation requires adding two additional days of incubation for every Vel below the standard forty, though Guild regulations strongly advise against allowing such fluctuations. Maintaining this precise thermal environment ensures the entire generation hatches simultaneously, preventing older nymphs from cannibalizing unhatched eggs.

1.2.2. The Crawling Nymph

Upon emerging from the egg, the insect enters the second stage of its life cycle as a Crawling Nymph. During

this phase, the insect resembles a segmented, pale caterpillar lacking any photogenic capabilities. The nymph's primary objective is rapid caloric consumption to build the biological reserves necessary for metamorphosis. Because they are terrestrial and cannot yet fly, nymphs require specialized feeding mechanisms. Sponge Troughs are installed along the Iron-Oak branches, soaked in a diluted nectar mixture that the nymphs can safely consume without the risk of drowning. The Crawling Nymph stage lasts for approximately three weeks, during which the insect will molt several times, shedding its restrictive outer exoskeleton to accommodate its rapidly expanding body mass. Towards the end of this phase, the nymph's diet must be heavily fortified with crushed moon-root to begin the synthesis of lumium within its developing internal glands. Once it reaches critical mass, the nymph secures itself to the underside of a sturdy leaf or branch and spins a thick, silken chrysalis, beginning the profound internal transformation required for the final stage of its life.

1.2.3. The Glowing Imago

The transition process from Nymph to Imago takes place entirely within the protective casing of the chrysalis. Over a period of ten days, the larval tissues break down into a nutrient-rich cellular soup before reforming into the complex adult structures. The most significant development during this transition is the formation of the lumium glands within the thorax and the growth of the delicate, expansive wings. When the metamorphosis is complete, the chrysalis splits, and the adult moth, now known as the Glowing Imago, emerges. The Imago pumps circulatory fluids into its wings to expand and dry them, a process that takes several hours. Once fully dried, the adult takes flight, its thorax igniting with the steady, pulsing light that defines the species. A healthy adult Glowing Imago produces an average output of fourteen Candons at peak photogenesis. The adult stage is dedicated entirely to breeding and light production. Unlike the nymphs, the imago feeds exclusively on fermented synthetic nectar from gravity drip dispensers. Proper management of the imago population ensures a constant, reliable harvest of ambient luminescence for the Guild.

1.3. Understanding Photogenesis in Moths

The most defining characteristic of the Lumen Moth is its ability to generate biological light, a phenomenon that has fascinated the Keepers Guild since its founding in Oakhaven. In the context of the Lumen Moth, photogenesis is defined as the continuous, enzymatically catalyzed production of visible light within the biological structures of the insect. Unlike the momentary flashes of common fireflies, the Lumen Moth produces a sustained, powerful glow designed to illuminate the dark valleys of the Alabaster Mountains. This process is not merely decorative; it is a fundamental metabolic function tied directly to the moth's respiratory and circulatory systems. The light serves to communicate with the swarm, attract mates, and ward off predators who are blinded by sudden flares of intensity. To sustain this continuous output, the moth relies heavily on its diet, specifically the intake of crushed moon-root, which provides the foundational chemical building blocks necessary for the reaction. Keepers must possess a deep understanding of the biochemistry involved to maintain a high yield of light during the harvest.

The chemical reaction that produces this light occurs entirely within the specialized thorax of the insect. The process is driven by the interaction between lumium, the organic compound synthesized from the moon-root, and atmospheric oxygen. As the moth breathes, oxygen is drawn in through the abdominal spiracles and pumped directly into the dense network of tracheal tubes surrounding the lumium glands. When the

atmospheric oxygen makes contact with the lumium, a rapid oxidation reaction takes place. This reaction releases energy not as heat, but as pure, cold light. The intensity of the light is directly proportional to the volume of oxygen introduced to the glands, which is why the light pulses in time with the moth's muscular respiratory contractions. This highly efficient chemical reaction between lumium and atmospheric oxygen results in an output of fourteen Candons per healthy adult. The lack of heat byproduct is crucial, as any thermal energy release would quickly cook the insect from the inside out.

Maintaining the stability of this chemical reaction requires constant vigilance from the Keeper, as dietary deficiencies will immediately manifest in the quality of the light. A healthy swarm emits a smooth, rhythmic pulse. However, if a Keeper observes a swarm where the light flickers erratically, they must swiftly diagnose the cause of this irregular pulsing. In almost all cases, irregular pulsing in an adult moth's light is caused by an iron deficiency in their diet. Iron acts as a crucial stabilizing catalyst in the oxidation of lumium. Without sufficient iron, the reaction stutters and misfires, resulting in chaotic, unpredictable flashes rather than a steady glow. To correct this, the Keeper must immediately supplement the synthetic nectar with iron-rich additives, often derived from the sap of the Iron-Oak tree. Failing to diagnose and treat this iron deficiency not only ruins the luminous harvest by producing low-quality light, but it also causes severe physical stress to the moths, ultimately shortening the lifespan of the Glowing Imago. A diligent Keeper will make a habit of standing in the center of the glass house during the darkest hours of the night, allowing their eyes to adjust to the ambient glow. By observing the collective rhythm of the swarm, one can detect the slightest erratic flicker against the canopy. This visual inspection is the most reliable method for monitoring the chemical stability of the photogenesis reaction across thousands of individual insects simultaneously.

1.4. Breeding Cycles and Seasonal Changes

Reproduction within the glass house is a carefully managed process that requires strict adherence to seasonal rhythms and environmental controls. The propagation of the swarm is cyclical, tied directly to the environmental conditions maintained by the Keepers. The duration of the standard breeding cycle for the Lumen Moth is exactly twenty-eight days. During this twenty-eight day period, the swarm progresses from the initial mating chase, through oviposition by the females, and concludes with the hatching of the dormant eggs into crawling nymphs. Keepers coordinate their nectar brewing, harvesting schedules, and structural maintenance around this rigid four-week cycle. By maintaining a constant internal temperature of forty-two Vels, the optimal ambient temperature for a healthy adult swarm, Keepers can ensure that this breeding cycle repeats continuously throughout most of the year. This perpetual cycle is essential for meeting the Guild's rigorous quotas for light production, as the constant influx of new imagoes replaces the older generation as their lumium glands naturally deplete.

However, even within the climate-controlled environment of a glass house, the moths remain deeply connected to the external seasonal shifts of their native Alabaster Mountains. The most significant of these seasonal changes is the arrival of the Frost Moon, an astronomical and meteorological event that occurs in the tenth month of the calendar. The effects of the Frost Moon on egg production are profound and unavoidable. As the Frost Moon rises, subtle changes in atmospheric pressure and ambient light spectra penetrate the glass panels, triggering an instinctual hibernation response within the swarm. The Frost Moon naturally halts all egg production for exactly thirty days. During this time, the adult moths enter a state of reduced metabolic activity, flying less frequently and consuming significantly less nectar. It is a time of rest for

the swarm. The Keeper's Code of Ethics explicitly protects this period of dormancy. Harvesting light from a dormant swarm during the Frost Moon violates Section Four of the Code, as the stress of the harvest could easily exhaust their depleted energy reserves and collapse the entire population.

Occasionally, external weather patterns or anomalies in the lunar cycle can cause a delayed seasonal shift, meaning the instinctual onset of the Frost Moon dormancy does not align with the calendar. When this occurs, the Keepers must proactively adjust breeding conditions to counteract a delayed seasonal shift. To manually induce the dormancy period, the Keeper must gradually lower the internal temperature of the glass house from the optimal forty-two Vels down to thirty-five Vels over the course of a week. Simultaneously, the nectar supply must be diluted, reducing the concentration of sugar-sap and crushed moon-root. This artificial cooling and dietary restriction successfully mimics the arrival of the Frost Moon, tricking the moths' physiology into halting egg production. Once the thirty-day rest period has elapsed, the Keeper will slowly raise the temperature back to forty-two Vels and restore the rich nectar supply, effectively restarting the standard twenty-eight day breeding cycle. Mastering these environmental adjustments separates a novice from a true Luminary of the Guild. Failure to manage these breeding cycles correctly can lead to catastrophic population crashes. If a Keeper ignores a delayed seasonal shift and attempts to force the swarm to continue breeding through the tenth month, the resulting eggs will often be sterile, and the adult moths will suffer from premature wing deterioration. Respecting the temporal boundaries set by the Frost Moon is not merely a matter of Guild ethics; it is a fundamental biological necessity for the long-term survival of the captive species.

1.5. Behavioral Patterns of the Swarm

The collective movement of thousands of Lumen Moths within the confined space of a glass house is not random; it is governed by distinct behavioral instincts. A Keeper must be able to read the mood and health of the swarm simply by observing their flight paths. There are four standard flight patterns of the swarm that every Keeper must commit to memory: the feeding drift, the mating chase, the resting cluster, and the canopy patrol. The feeding drift is a slow, undulating movement near the lower branches where the gravity drip dispensers are located. The mating chase is characterized by rapid, erratic darting as males pursue the pheromone trails of females. The resting cluster involves thousands of moths clinging motionless to the Iron-Oak frames, pulsing their light in unison. Finally, the canopy patrol is a steady, circular flight path just beneath the prismatic glass panels, where the moths gather to absorb residual heat. These four patterns represent a healthy, balanced environment and indicate that the swarm's needs are being met.

Deviations from these standard patterns serve as immediate warning signs of environmental or dietary distress. One of the most alarming abnormal behaviors is the spiral descent. When a Keeper observes thousands of moths flying in tight, downward corkscrews toward the floor of the glass house, they must act quickly. Interpreting the meaning of the spiral descent behavior is straightforward: it means the swarm is starved of moon-root. This specific nutrient deficiency disrupts their spatial orientation and buoyancy. Without the chemical energy provided by the moon-root, their wing muscles spasm, causing the distinctive downward spiral. If this behavior is observed, the Keeper must immediately inspect the nectar vats and restock the feeding troughs with freshly crushed moon-root before the insects exhaust themselves and perish on the floor.

Another critical behavioral anomaly is swarm aggregation in the upper corners of the glass house. This occurs when the moths suddenly abandon the central canopy and pack themselves tightly into the highest structural angles, desperately pressing against the glass. A Keeper must formulate a response plan for swarm aggregation in the upper corners, which is triggered by a sudden internal pressure drop of more than five Torr-Pascs per hour. This rapid depressurization simulates the approach of a severe mountain storm. To resolve this, the Keeper must immediately operate the manual Draft-Bellows to equalize the internal air pressure, pumping air into the chamber until the pressure stabilizes and the moths safely disperse back into the canopy patrol pattern.

Temperature control is equally vital to maintaining stable swarm behavior. The optimal temperature is forty-two Vels, but extreme heat is highly dangerous. If the internal climate reaches fifty-five Vels, the moths will experience sudden swarm panic. During a swarm panic, the insects fly blindly at maximum speed, ricocheting off the glass panels and each other. This frenzied state leads to rapid mass exhaustion and can cause permanent damage to their delicate wings. Keepers must ensure the cooling vents are properly managed at dusk to prevent the heat of the day from becoming trapped. Understanding and reacting to these behavioral patterns is the essence of moth keeping, transforming a chaotic cloud of insects into a harmonious and productive source of light.

2. Glass House Architecture and Climate

Learning objectives

LO-2.1.1 Identify the three approved building materials for Glass House frames. (K1)

LO-2.1.2 Explain why Iron-Oak is preferred over steel for structural supports. (K2)

LO-2.1.3 Assess a glass house structural blueprint for load-bearing flaws. (K3)

LO-2.2.1 List the standard locations for cross-breeze vents. (K1)

LO-2.2.2 Explain how draft control impacts moth flight stability. (K2)

LO-2.2.3 Operate the Draft-Bellows to equalize pressure during a storm. (K3)

LO-2.3.1 Describe the purpose of prismatic panels in the canopy. (K2)

LO-2.3.2 Explain how Lunar-Tinted Glass filters harmful solar radiation. (K2)

LO-2.3.3 Select the appropriate glazing replacement for a shattered south-facing panel. (K3)

LO-2.4.1 State the target temperature range in Vels for dusk transition. (K1)

LO-2.4.2 Explain why a Glass House is cooled at dusk rather than dawn. (K2)

LO-2.4.3 Calibrate a Vels-Thermometer using the freezing point of synthetic nectar. (K3)

LO-2.5.1 Define the optimal humidity percentage for adult moth longevity. (K1)

LO-2.5.2 Describe the interaction between humidity and photogenesis intensity. (K2)

LO-2.5.3 Troubleshoot a failing misting system using pressure logs. (K3)

Constructing and maintaining a proper enclosure is fundamental to the success of any swarm kept outside the Alabaster Mountains. This chapter examines the strict architectural codes enforced by the Keepers Guild of Oakhaven, detailing the materials, ventilation mechanisms, specialized glazing, and climate control techniques required to sustain a productive artificial habitat.

2.1. Structural Requirements of a Glass House

Every sanctuary built to house Lepidoptera luminis must adhere to the strict architectural mandates established in Oakhaven. When erecting the skeleton of an enclosure, builders may select only from a restricted list of certified components. The three Guild-certified structural materials for constructing these enclosures are cured Iron-Oak, quarried Alabaster-Stone, and spun Copper-Resin. Alabaster-Stone is utilized exclusively for the subterranean foundation walls and perimeter footings, providing an immovable anchor against shifting mountain soils. Spun Copper-Resin is molded into the secondary glazing ribs, window muntins, and light-weight archways that hold individual panes in place without adding excessive overhead mass. However, when it comes to the primary load-bearing pillars, central ridge beams, and vaulted trusses, Iron-Oak is the sole wood permitted by the Guild. No other timber or refined metal may bear the weight of the canopy. In fact, during an official inspection, the unauthorized incorporation of steel in frame construction results in the immediate demotion of the responsible Keeper down to the rank of Apprentice.

Novice builders often wonder why modern metallurgy is strictly forbidden in the upper skeleton in favor of

harvested timber. Cured Iron-Oak is favored over refined steel for heavy load-bearing pillars for three critical biological and physical reasons. First, steel conducts external heat and cold far too rapidly, creating localized thermal bridges that cause condensation and shock Dormant Eggs laid near the frame. Iron-Oak possesses natural cellular insulation that maintains a stable surface warmth near forty Vels. Second, refined steel generates subtle magnetic eddies when exposed to mountain wind friction, which disorients the delicate sensory antennae of the males during the mating chase. Third, Iron-Oak bark retains trace mineral oils that naturally repel the Glass-Borer Beetle, whereas steel joints offer smooth, warm crevices where beetle larvae thrive before chewing through adjacent pane seals. Furthermore, Iron-Oak absorbs the high-frequency acoustic vibrations of a pulsing swarm, whereas steel amplifies those vibrations into a resonant hum that agitates the resting clusters.

Before a single stone is laid, a senior Keeper must review the architectural drawings to verify that the proposed building will not collapse under heavy snow or mountain gales. To evaluate an enclosure plan for weight-distribution hazards, the reviewer must apply three numeric standards established by the Guild engineers. First, primary Iron-Oak pillars must be spaced no more than twelve feet apart along the north and south load walls; any span exceeding twelve feet will cause the overhead Copper-Resin ribs to sag and pinch the glazing. Second, the main roof pitch must rise at a steep angle between thirty-five and forty-five degrees so that winter ice sheets slide off rather than crushing the canopy. Third, every central ridge junction where four or more Copper-Resin ribs meet must be supported directly from below by a vertical Iron-Oak post anchored into an Alabaster-Stone pier, never cantilevered over open floor space. If a blueprint displays a sixteen-foot gap between pillars, a flat twenty-degree roof slope, or an unsupported central rib junction, the reviewer must reject the design and mandate immediate structural reinforcement before construction begins.

2.2. The Vents and Draft Control

Airflow within a glass house must replicate the gentle, predictable currents of the deep valleys in the Alabaster Mountains. Stagnant air suffocates the lumium oxidation reaction in the moths' thoraxes, while violent gusts batter their delicate wings against the timber supports. To achieve a balanced atmosphere, builders install a coordinated network of airflow openings at three precise architectural tiers. The standard placements for these airflow openings are the low north-wall intake louvers positioned six inches above the Alabaster-Stone floor, the mid-level east-west baffle slots situated eight feet up the side walls, and the high apex exhaust ridge running along the central Iron-Oak spine directly beneath the highest canopy panels. By opening the low north louvers and the high apex ridge simultaneously, a Keeper establishes a gentle thermal chimney that draws cool, oxygen-rich air across the lower feeding troughs and vents warm, stale air out the top.

Careful management of these openings is essential because interior wind speed directly governs how the insects navigate the enclosure. Moving air currents dictate whether the swarm can hold its four routine flight formations without expending excess caloric reserves. Because a Glowing Imago possesses an exceptionally high wing-surface-to-body-mass ratio, any internal breeze exceeding three miles per hour overpowers their muscular downstrokes. When drafts are properly baffled below this threshold, moths glide effortlessly along the thermal chimney, conserving energy for photogenesis and keeping their needle-like or feathery antennae oriented toward pheromone trails. Conversely, if an unbaffled cross-wind shears through the mid-level slots, it disrupts the slow feeding drift near the gravity drip dispensers and scatters the high canopy patrol into

chaotic, exhausting tumbles. Turbulent eddies also strip away the microscopic boundary layer of warm air clinging to the insect's thorax, forcing the moth to burn moon-root reserves for warmth rather than light.

Normal louver adjustments suffice during calm weather, but severe alpine squalls require active mechanical intervention. When a storm front sweeps over the glass house, barometric pressure outside plummets, sucking air out through microscopic pane seams. As established in biological studies, an internal pressure drop exceeding five Torr-Pascs per hour triggers swarm aggregation in the upper corners. To prevent this using the manual Draft-Bellows, the Keeper must follow a strict four-step pneumatic sequence. First, the Keeper latches the mid-level baffle slots and the high apex exhaust ridge completely shut while leaving the low north intake louvers cracked open by one inch. Second, the Keeper connects the leather nozzle of the Draft-Bellows to the brass check-valve embedded in the main north pillar. Third, watching the wall-mounted Torr-Pasc gauge, the Keeper pumps the wooden handles at a steady cadence of twenty strokes per minute, forcing compressed air through the floor ducts until internal resistance climbs back to the standard baseline of one hundred and forty Torr-Pascs. Fourth, once the storm passes and external pressure stabilizes, the Keeper bleeds excess pressure by unlatching the mid-level baffles before slowly reopening the apex ridge.

2.3. Glass Glazing and Refraction

2.3.1. Prismatic Panels

The transparent skin of the enclosure does far more than trap warmth; it shapes the electromagnetic environment of the swarm. The upper vault of the structure is glazed entirely with thick, wedge-cut panes known as prismatic panels. Installed at the highest angles of the roof, these multi-faceted panes serve two vital optical functions for the captive insects. First, during the daytime hours, the angled quartz ridges split incoming harsh mountain sunlight into diffuse, low-energy bands of violet and indigo light that spread evenly across the interior, eliminating sharp shadows and hot spots that would otherwise burn the resting clusters clinging to the Iron-Oak beams. Second, after nightfall, the interior facets of the panels act as total-internal-reflection mirrors, capturing upward-escaping bioluminescence from the canopy patrol and bouncing that cold light back down toward the lower branches. This downward reflection amplifies ambient illumination around the feeding dispensers by nearly thirty percent and stimulates synchronized pulsing across the entire swarm. To maintain this optical clarity, Keepers wash the interior facets monthly using diluted V-Scrub solution, taking care to rinse every drop before moths return to the canopy.

2.3.2. Lunar-Tinted Glass

While the upper roof utilizes clear prismatic quartz, the vertical and sloped lower walls require a very different optical barrier. Direct, unfiltered midday sun striking a moth's translucent thorax will rapidly photobleach the lumium reserves inside its glands and sear its compound eyes. To prevent this cellular damage, Guild codes mandate a specialized glazing known as Lunar-Tinted Glass for all sun-exposed exposures. During the molten pouring process in the furnaces of Oakhaven, master glaziers infuse the silica batch with precise trace amounts of metallic silver and powdered cobalt. As the pane cools, the suspended silver particles form a microscopic lattice within the sheet. When fierce afternoon sunlight strikes the exterior of the pane, this silver lattice reflects and scatters ninety-five percent of destructive ultraviolet and high-thermal infrared wavelengths back into the sky, while permitting only cool, gentle silver-blue wavelengths to pass into the chamber. This filtered light closely mimics the natural twilight of the deep Alabaster Mountain gorges, keeping the swarm

calm and preserving their delicate lumium chemistry throughout the brightest summer afternoons.

2.3.3. Glazing Selection and Replacement

Hailstorms and falling mountain branches frequently crack individual panes, requiring the Keeper to act swiftly as a glazier. Selecting the correct replacement sheet from the storage shed depends entirely on the cardinal orientation and elevation of the damaged frame section. According to Guild architectural rules, Lunar-Tinted Glass containing trace silver must be installed on all south-facing panels, where solar exposure is most intense and prolonged. North-facing vertical walls, which never receive direct sun, are fitted with standard double-thick clear soda-lime panes to maximize ambient daylight during winter, while the upper roof vault above the twelve-foot collar beam exclusively requires wedge-cut prismatic panels. When replacing a shattered south-facing pane, the Keeper first clears the jagged shards from the spun Copper-Resin muntins using heavy leather gauntlets. Next, the Keeper selects a certified sheet of silver-infused Lunar-Tinted Glass, verifies its hallmark stamp in the lower corner, and beds the edges into fresh Copper-Resin putty with the slightly concave silvered face oriented toward the exterior sun. Installing clear glass or a prismatic panel on a south wall is a severe safety violation that will blind the swarm by noon.

2.4. Temperature Management at Dusk

Thermal stability is the heartbeat of a productive glass house. As established in foundational biology, a healthy adult swarm thrives at an optimal ambient climate of exactly forty-two Vels, while Dormant Eggs require a steady forty Vels and a climb to fifty-five Vels triggers catastrophic swarm panic. The most perilous period of the daily thermal cycle occurs at twilight, when daytime solar gain lingers in the Alabaster-Stone foundations just as the swarm wakes from its resting clusters to begin high-energy nocturnal flight. During this one-hour dusk transition window, Keepers must actively guide the interior climate through a narrow, cooling corridor running from forty-four Vels at sunset down to precisely forty-one Vels by full dark, before letting the night furnaces settle the chamber at the standard forty-two Vels. Holding the dusk transition within this forty-one to forty-four Vel bracket prevents residual afternoon warmth from combining with the rising metabolic activity of waking moths.

Apprentices often ask why the enclosure is deliberately cooled as night approaches rather than at dawn when the sun rises. This evening cooling protocol is dictated by the interaction between the enclosure's thermal mass and the biological rhythms of Lepidoptera luminis. Throughout the afternoon, even behind Lunar-Tinted Glass, the heavy Alabaster-Stone floor and Iron-Oak pillars absorb solar energy and radiate stored heat well into the evening. Simultaneously, at dusk, thousands of Glowing Imagoes transition from motionless resting clusters into the rapid mating chase and canopy patrol. Although the chemical oxidation of lumium itself produces cold light, the vigorous muscular contractions of ten thousand pairs of wings generate substantial biological heat and consume massive quantities of oxygen. If a Keeper waited until dawn to vent the house, the combination of radiating stone heat and sudden dusk wing-friction would spike the interior past fifty-five Vels within thirty minutes of sunset, triggering immediate swarm panic. At dawn, by contrast, the moths settle into motionless clusters and external mountain air is at its coldest, requiring heat retention rather than cooling.

Because a two-Vel error during dusk transition can ruin a breeding cycle, Keepers must verify the accuracy of their brass Vels-Thermometers every week. The standard Guild method relies on the immutable physical

property that pure synthetic nectar freezes at exactly twelve Vels. To execute this calibration, the Keeper fills a small insulated copper beaker with ten Drams of unfermented standard synthetic nectar and places the beaker inside an ice-and-salt bath. The Keeper submerges the bulb of the Vels-Thermometer into the nectar while stirring gently with a glass rod. At the exact moment translucent slush crystals begin to form across the surface of the nectar, the liquid is at its twelve-Vel freezing point. The Keeper immediately inspects the mercury column; if the indicator does not rest precisely on the twelve-Vel hash mark, the Keeper turns the brass calibration screw at the base of the tube until the pointer aligns with twelve Vels, locking the nut before returning the instrument to the central Iron-Oak pillar.

2.5. Humidity Regulation Systems

Airborne moisture is the invisible partner to temperature in the glass house climate. While wild moths in the Alabaster Mountains rely on glacial waterfalls and valley fog to keep their bodies hydrated, captive swarms depend entirely on mechanical misting networks running along the Iron-Oak trusses. For maximum Glowing Imago lifespan and wing flexibility, the relative atmospheric moisture inside the enclosure must be held within a strict optimal band between sixty-five and seventy percent. If relative moisture falls below sixty-five percent, the microscopic chitin scales on the wings desiccate and flake off, shortening the adult lifespan from four weeks to less than ten days. Conversely, if moisture climbs above seventy percent for more than six hours, condensation beads form on the feathery antennae of the males, weighing them down and blinding their pheromone receptors, while simultaneously encouraging toxic mold growth inside the gravity drip dispensers.

Beyond preserving wing flexibility, ambient moisture directly governs the brightness of the swarm's bioluminescence. As detailed in chapter one, photogenesis depends on drawing atmospheric oxygen through the abdominal spiracles and pumping it through tracheal tubes into the thorax's lumium glands. The internal linings of these microscopic tracheal tubes must remain coated in a thin film of moisture so oxygen molecules can dissolve and diffuse across the gland membranes. When relative humidity sits in the ideal sixty-five to seventy percent range, gas exchange operates at peak efficiency and a healthy adult effortlessly emits its full fourteen Candons. If the air dries out below fifty percent, the tracheal linings constrict and crust over to prevent internal dehydration; this defensive constriction starves the lumium glands of oxygen, causing the swarm's output to dim to five or six Candons. Conversely, if humidity exceeds eighty percent, microscopic water droplets clog the external spiracle valves, choking the oxygen supply and causing the light to sputter and dim just as severely.

Maintaining this delicate moisture balance requires daily inspection of the overhead brass misting nozzles and their supply pipes. Keepers monitor the system by cross-referencing the wall hygrometer against the hydraulic pressure logs recorded at the main water pump. Under normal operation, the pump maintains a steady line pressure of ninety Torr-Pascs, delivering a fine, invisible vapor through the nozzles every twenty minutes. When a misting system fails, the Keeper diagnoses the exact fault by comparing the pressure log trend against the humidity reading. If the log shows line pressure slowly dropping below sixty Torr-Pascs while chamber humidity falls below fifty-five percent, the fault is a split copper supply pipe or a failing pump leather near the reservoir. However, if the log shows line pressure spiking above one hundred and twenty Torr-Pascs while chamber humidity simultaneously plummets, the pump is working against a blockage: sugar-sap residue or mineral scale has crystallized inside the overhead brass nozzle tips. In that scenario, the Keeper must shut the main valve, unscrew the clogged nozzle heads, and soak them in V-Scrub solution

for ten minutes before flushing the line.

3. Feeding and Nectar Preparation

Learning objectives

LO-3.1.1 Name the five core ingredients of synthetic nectar. (K1)

LO-3.1.2 Explain the role of crushed moon-root in the moth's diet. (K2)

LO-3.1.3 Calculate the correct ratio of sugar-sap to water for a swarm of 500. (K3)

LO-3.2.1 Outline the sequence of steps for the nectar brewing process. (K1)

LO-3.2.2 Discuss the importance of temperature control during brewing. (K2)

LO-3.2.3 Modify a brewing recipe to account for high ambient humidity. (K3)

LO-3.3.1 State the minimum fermentation time required for standard nectar. (K1)

LO-3.3.2 Explain the chemical changes that occur during the resting period. (K2)

LO-3.3.3 Identify signs of over-fermentation in a resting vat. (K2)

LO-3.4.1 Name the two primary types of feeding dispensers. (K1)

LO-3.4.2 Compare the efficiency of Gravity Drip Dispensers versus Sponge Troughs. (K2)

LO-3.4.3 Install a Sponge Trough in a high-traffic feeding zone. (K3)

LO-3.5.1 Identify the visual symptoms of moon-root deficiency. (K1)

LO-3.5.2 Explain how iron deficiency impacts the color of the moth's light. (K2)

LO-3.5.3 Devise a diet plan to rehabilitate a nutrient-deficient swarm. (K3)

Providing artificial nourishment is among the most demanding responsibilities entrusted to a Keeper. Because wild flora of the Alabaster Mountains cannot easily take root under glass, captive swarms rely entirely upon meticulously prepared liquid diets. This chapter covers the selection of raw compounds, the chemistry of brewing and fermentation, the deployment of feeding stations, and the diagnostic methods used to identify nutritional imbalances before colony health declines.

3.1. Ingredients of Synthetic Nectar

Maintaining a thriving colony under glass demands an unyielding commitment to dietary quality. Keepers must prepare a uniform liquid feed that replicates the rich floral exudates of the Alabaster heights. The recipe established by the Guild of Oakhaven relies on five essential components: crushed moon-root, filtered mountain spring water, raw sugar-sap harvested from Iron-Oak bark, powdered copper-flower petals, and mineral sea-ash. Water serves as the neutral fluid medium, while copper-flower powder supplies trace metallic salts that strengthen the chitinous veins supporting the wings. Mineral sea-ash introduces alkaline buffers that keep the overall solution stable, protecting the insects' mouthparts from corrosive acidity. Each ingredient must be weighed using brass balance scales before entering the copper brewing vats, as even minor variances in dry proportions disrupt the digestive tracts of the insects.

Among these components, crushed moon-root is the irreplaceable cornerstone of the creature's sustenance.

This subterranean tuber contains high concentrations of precursor alkaloid proteins that the insect cannot synthesize on its own. Once ingested through the long proboscis, the digestible starches of the tuber are broken down in the midgut and absorbed into the hemolymph, where they travel directly to the central thoracic cavity. Inside the lumium glands, these alkaloids undergo enzymatic transformation into pure lumium, the vital compound that oxidizes with atmospheric oxygen to generate luminescence. Depriving an adult of this tuber completely cuts off its internal synthesis pathway, exhausting existing lumium stores within days and extinguishing the biological glow.

Calculating batch ratios requires adjusting the fluid proportions based on the size of the adult population. For every standard cluster of five hundred Glowing Imagoes, the baseline daily ration requires exactly two hundred Drams of thick sugar-sap blended with six hundred Drams of spring water, establishing a strict one-to-three volumetric ratio of sap to water. If a Keeper manages an intermediate colony size, such as two hundred and fifty adults, the volume is halved to one hundred Drams of sap and three hundred Drams of water, maintaining that exact one-to-three relationship. Increasing the sugar-sap proportion beyond this limit thickens the fluid, clogging the narrow siphon channels of the mouthparts, whereas excessive dilution deprives the insects of the energy needed to power continuous wing beats.

3.2. The Brewing Process

Preparing synthetic nectar is an exact culinary science governed by rigorous Guild protocols. The production follows an unbroken sequence of five physical actions: pre-soaking the moon-root, heating the spring water base, dissolving the sugar-sap and dry minerals, infusing the strained moon-root liquid, and cooling the mixture. Before any heat is applied, crushed moon-root must be soaked in cold spring water for exactly twelve hours in unglazed earthenware jars. This extended steeping extracts the active alkaloids without destroying their heat-sensitive molecular rings. Once the twelve hours elapse, the Keeper strains the steeped liquid through static-free silk, discarding the coarse fibrous pulp.

Thermal regulation throughout the cooking phase dictates the chemical viability of the batch. The strained water base is brought into a heavy copper kettle and heated to thirty-five Vels before adding two hundred Drams of sugar-sap and the measured copper-flower and mineral sea-ash powders. The liquid must never exceed forty Vels during simmering. Allowing the brewing vat to spike above forty Vels permanently denatures the delicate moon-root alkaloids, stripping them of their photogenic properties and rendering the entire batch useless. Conversely, if the kettle fails to reach thirty Vels, the raw sap fails to dissolve completely, settling into a dense sludge at the base of the kettle that starves feeding adults of necessary carbohydrates. A wooden paddle is turned continuously in slow clockwise strokes to prevent localized hot spots along the bottom of the vat.

When environmental hygrometers record high ambient moisture in the surrounding air, the brewing recipe must be adapted to safeguard the fluid. Elevated atmospheric moisture reduces evaporation from open dispensers, causing the nectar to absorb excess moisture and thin out prematurely in the glass house. Under conditions where relative humidity exceeds seventy percent, the Keeper modifies the formula by increasing the sugar-sap volume by ten percent - raising it to two hundred and twenty Drams per six hundred Drams of water - and reducing the initial boiling volume of spring water by fifty Drams. This produces a slightly denser starting nectar that resists humidity-induced dilution and maintains its caloric value over long feeding shifts.

3.3. Fermentation and Resting Periods

Raw, freshly brewed nectar cannot be dispensed directly to a swarm. The solution requires a stabilization phase during which wild airborne yeasts from the Alabaster heights interact with the dissolved wood sugars. Standard nectar requires exactly forty-eight hours of resting time to ferment properly in sealed oak casks kept at thirty-two Vels. Distributing the fluid before this forty-eight-hour resting period elapses results in severe gastrointestinal distress among adult moths, as the unfermented sugars cause lethal bloating within the segmented abdomen. Keepers mark the bung of each cask with chalk, noting the precise hour the vat was sealed to avoid tapping an immature batch.

Throughout this mandatory two-day resting window, profound chemical changes take place inside the vat. Complex polysaccharide chains present in the raw Iron-Oak sap undergo enzymatic breakdown, converting into simple, digestible monosaccharides and trace alcohols. Concurrently, the dissolved moon-root alkaloids bind with these light alcohols, forming a shelf-stable organo-luminescent solution. This binding process ensures that when the fluid enters the moth's hemolymph, the lumium precursors release smoothly and evenly into the thoracic glands over an extended twelve-hour digestion cycle, rather than flooding the system all at once. The pH of the liquid gently drops from neutral to slightly acidic, which acts as a natural preservative against bacterial contamination.

Keepers must remain vigilant against the perils of over-fermentation in resting vats. If a cask remains unsealed or is stored above thirty-four Vels, the yeast culture consumes all available monosaccharides and begins decomposing the bound moon-root alkaloids. Over-fermented nectar produces a toxic gas that can corrode the iron bands on gravity drip dispensers. Detecting this degradation early is essential for colony survival. The primary indicator of over-fermented nectar is a pungent, sharp odor resembling burnt almonds, paired with a frothy, gray scum accumulating across the top of the liquid. Any cask displaying this burnt-almond aroma must be discarded immediately; under no circumstances may it be fed to insects, as ingestion triggers fatal metabolic collapse.

3.4. Feeding Dispensers and Troughs

3.4.1. Gravity Drip Dispensers

Delivering prepared feed across a glass house requires matching the dispensing apparatus to the life stage of the insects. The two primary types of feeding dispensers approved by the Guild are Gravity Drip Dispensers and Sponge Troughs. Gravity Drip Dispensers consist of an inverted cylindrical glass reservoir attached to a brass siphon tube with a microscopic droplet valve. These units are suspended from overhead Iron-Oak trusses, spaced fifteen feet apart throughout the upper third of the canopy. The valve releases a measured droplet of fermented nectar every sixty seconds, perfectly suited for the hovering feeding drift of adult Glowing Imagoes. The enclosed glass reservoir prevents dust and debris from settling into the liquid, ensuring that large adult populations receive a sterile, continuous supply of nourishment without manual refilling during peak harvest nights.

3.4.2. Sponge Troughs

In contrast to the elevated glass drippers, Sponge Troughs are open wooden trays lined with porous, natural

sea sponges that absorb and hold liquid nectar without creating standing pools. Comparing the operational characteristics of the two systems reveals their distinct developmental purposes. Gravity Drip Dispensers boast high volumetric efficiency for flying adults, dispensing twenty Drams of fluid per hour without human intervention, but their dripping valves present a fatal drowning hazard for wingless Crawling Nymphs. Sponge Troughs, while requiring manual re-soaking every eight hours, offer a stable, slip-free terrestrial feeding surface where flightless nymphs can feed safely without any risk of becoming trapped or submerged in fluid.

3.4.3. Installing Troughs in High-Traffic Zones

Securing a Sponge Trough in an active nursery corridor requires careful physical installation to prevent dislodgement under the weight of hundreds of nymphs. The Keeper must first select a horizontal Iron-Oak branch located within three feet of the Alabaster-Stone floor, directly adjacent to the primary hatching barks. The wooden trough is mounted to the timber using two non-reactive copper brackets, driven into the underside of the limb with brass pins. The Keeper tests the fixture by applying two pounds of downward hand pressure to verify that the trough will not tilt. Once secured, the natural sea sponge is soaked in one hundred Drams of prepared nymph nectar, squeezed to remove air pockets, and pressed firmly into the wooden bed so its upper surface sits flush with the lip of the trough, granting young crawlers effortless access from the surrounding bark.

3.5. Nutritional Deficiencies

The visual appearance and flight patterns of a swarm serve as a living gauge of dietary balance. When a batch of nectar is mixed improperly, the physical consequences manifest across the colony within forty-eight hours. The visual symptoms of moon-root deficiency are striking and unmistakable. Deprived of sufficient tuber alkaloids, the insects can no longer sustain the muscular buoyancy required for steady flight. The moths exhibit the distinctive spiral descent, flying in helpless downward corkscrews until they crash onto the lower foliage. Simultaneously, their illumination dims from the standard fourteen Candons down to a dull, sickly glow below three Candons, while their segmented abdomens appear hollow and shriveled under examination.

A different dietary imbalance arises when mineral levels fall out of equilibrium, particularly the vital iron compounds supplied by Iron-Oak sap. While iron deficiency in a moth's diet causes their light to flicker erratically rather than pulse smoothly, it also triggers a dramatic chromatic aberration in the luminescence itself. The normal, cool silver-blue beam shifts into an unstable, greenish hue as the enzymatic oxidation reaction misfires. Because the lumium molecule cannot form a stable electronic shell without iron ions acting as structural anchors, the wavelength of the emitted photons lengthens. A novice Keeper who ignores this greenish flicker risks widespread cellular degeneration across the thoracic tissues of the swarm.

Restoring health to a nutrient-compromised colony demands a structured rehabilitation schedule implemented over seven consecutive days. On day one and two, the Keeper drains all overhead gravity drip dispensers and cleans the glass reservoirs using diluted V-Scrub solution to eliminate all traces of the substandard feed. On days three through five, the swarm is placed on an emergency enrichment formula: the proportion of crushed moon-root is doubled, soaking for sixteen hours rather than twelve, while the liquid base is infused with thirty Drams of concentrated, iron-rich Iron-Oak sap per batch. Feeding is supplemented using floor-level Sponge Troughs to accommodate exhausted insects unable to fly to the canopy. On days six

and seven, the Keeper monitors the swarm with a Candon-meter, checking that output climbs back toward fourteen Candons and pulses return to a rhythmic silver-blue before standard feeding routines resume.

4. The Luminous Harvest and Regulation

Learning objectives

LO-4.1.1 List the tools required for the Luminous Harvest. (K1)

LO-4.1.2 Describe the sweep method used to collect ambient luminescence. (K2)

LO-4.1.3 Execute a mock harvest sweep without disrupting moth flight paths. (K3)

LO-4.2.1 State the maximum capacity of a standard Lumium Flask. (K1)

LO-4.2.2 Explain why harvested light must be stored in vacuum-sealed containers. (K2)

LO-4.2.3 Troubleshoot a leaking Lumium Flask valve. (K3)

LO-4.3.1 Define the Candon as a unit of light measurement. (K1)

LO-4.3.2 Explain how a Candon-meter reads ambient light saturation. (K2)

LO-4.3.3 Calculate the total Candon yield of a harvest using meter readings and volume. (K3)

LO-4.4.1 Recall the mandatory protective gear for harvesting. (K1)

LO-4.4.2 Explain the risk of light-burn to unprotected retinas. (K2)

LO-4.4.3 Apply emergency first aid for acute light-burn exposure. (K3)

LO-4.5.1 Name the three processing phases before transport. (K1)

LO-4.5.2 Discuss the impact of vibration on stored luminescence during transit. (K2)

LO-4.5.3 Prepare a transport manifest for a shipment of Grade A Lumium. (K3)

The culmination of a Keeper's labor is the extraction of pure bioluminescent radiance from the glass house canopy. Gathering this ethereal substance requires precision, specialized implements, and strict adherence to guild safety codes. This chapter details the technical procedures for sweeping ambient luminescence, sealing the fluid energy into vessels, calculating yield, protecting human sight, and preparing shipments for the lanterns of Oakhaven.

4.1. The Luminous Harvest Process

4.1.1. Preparing the Harvesting Nets

Gathering cold light from an active colony requires specialized mechanical gear designed to capture ethereal radiance without harming the fragile insects. A certified team must assemble three distinct pieces of apparatus before mounting the harvest scaffolding: the Lunar-Net, an evacuated copper collector manifold, and a portable pneumatic transfer siphon. The Lunar-Net itself is crafted from static-resistant silk strands woven with trace silver threads, attached to an extendable Iron-Oak handle. Before ascending to the upper canopy walkways, the Keeper must inspect the silk mesh under a magnifying glass to confirm there are no snagged fibers or frayed knots that could trap an insect's antennae. The metal intake collar at the net's base connects directly to the copper manifold, which channels captured airborne luminescence toward the storage

canisters through a flexible, light-tight hose.

4.1.2. The Sweep Method

Collecting the glowing aura shed by thousands of flying adults relies on smooth, fluid kinetic motion rather than aggressive netting. The harvest crew ascends to elevated wooden catwalks positioned high in the glass house vault. Guild regulations specify that the collector must execute wide, continuous figure-eight patterns precisely three feet below the canopy vents. Moving the net in these rhythmic figure-eights creates a soft low-pressure wake that gathers airborne luminescent vapors while gently nudging flying adults aside without touching their bodies. The collector must maintain a rigid vertical buffer; using the Sweep Method too close to the prismatic panels risks cracking the glass due to static buildup in the net. Keepers must preserve that exact three-foot clearance beneath the ceiling, ensuring the silk frame never brushes the multifaceted quartz overhead.

4.1.3. Practical Sweep Execution

Executing an effective harvest sweep requires an operator to synchronize their movements with the natural rhythm of the swarm's canopy patrol. When standing on the high walkways, the harvester holds the Lunar-Net with both hands, extending the Iron-Oak shaft at a forty-five-degree angle into the upper air currents. As the insects circle in their routine nocturnal pattern, the harvester sweeps the hoop through the air in wide, looping motions at a deliberate pace of one complete figure-eight every six seconds. If a moth drifts toward the hoop, the harvester must avoid jerking the pole backward; instead, they slowly tilt the net's rim parallel to the airflow, allowing the insect's own wing wake to glide it past the silk boundary. By maintaining smooth, unhurried arcs exactly three feet beneath the vents, the operator pulls concentrated light into the vacuum collar while the surrounding insects continue their circular patrol undisturbed.

4.2. Storing the Luminescence

Once captured by the intake collar, cold luminescence condenses into a heavy, glowing fluid that must be secured inside standardized guild glassware. The approved vessel for this material is the standard Lumium Flask, a double-walled borosilicate container encased in an outer brass cage. Every standard Lumium Flask holds exactly fifty Drams of condensed luminescence when filled to the interior graduation line. Filling a flask beyond this fifty-Dram mark prevents the neck valve from sealing flush, creating micro-fissures in the collar that permit pressure leakage. Apprentices must always measure fluid levels against the etched fifty-Dram boundary mark before disengaging the transfer siphon.

Containing this fluid energy requires absolute isolation from ambient air. Harvested light must be held in vacuum-sealed containers because lingering atmospheric oxygen causes uncontrolled, rapid degradation of the condensed compound. Inside the insect's thorax, the oxidation of lumium produces a steady, metered output of fourteen Candons under tight biological control. Outside the body, however, uninhibited contact with atmospheric oxygen triggers a runaway reaction, burning through the chemical energy within hours and leaving behind an inert, dark amber fluid. Evacuating all air from the vessel halts this oxidation, locking the substance in a state of suspended luminescence that remains stable for months until unsealed in municipal streetlamps.

During warehouse inspections, a Keeper will occasionally encounter a vessel exhibiting pressure loss or

minor leakage around its collar. To correct a leaking Lumium Flask valve, the Keeper must apply a strict maintenance sequence using certified materials. First, the Keeper moves the weeping flask into an unlit cooling chamber held at thirty-five Vels to slow liquid volatility. Second, the Keeper removes the outer brass collar clamp and cleans the exposed threaded fitting using an absorbent linen cloth dampened with spring water. Third, the Keeper inspects the seal and applies a thin, uniform ribbon of fresh resin derived from the same tree as the sugar-sap used in nectar, running the bead completely around the valve seat. Finally, the brass collar is re-seated and tightened with a torque wrench to fifteen inch-pounds, resealing the vacuum barrier before condensation escapes.

4.3. Measuring Light Yield in Candons

Quantifying the harvest output requires objective, standardized photometric units. The Keepers Guild defines the Candon as the fundamental unit of light measurement, representing the exact luminous flux emitted by a standard reference candle cast from refined mountain tallow. In practical terms, one healthy adult moth at peak photogenesis produces approximately fourteen Candons of illumination. A glass house containing ten thousand mature adults will illuminate the surrounding valley with several thousand Candons of radiance, which harvesters draw down into flasks during the dark hours following the dusk transition.

Measuring this radiant energy in the field is accomplished with an optical instrument called the Candon-meter. The device consists of an oblong brass housing containing a light-sensitive selenium sensor plate, a quartz diffusion lens, and an analog needle dial calibrated across a scale from zero to one hundred Candons. When a Keeper holds the meter toward the canopy, incoming photons pass through the diffusion lens and strike the selenium plate, generating a tiny electric current proportional to the light's intensity. This current moves the needle across the dial face, providing an instant readout of ambient light saturation within the chamber. Keepers must remember that the Candon-meter requires daily recalibration using the freezing point of synthetic nectar, which freezes at exactly twelve Vels, verifying that the mechanical pointer rests at true zero when placed in the chilled testing box.

Calculating the overall commercial yield of a harvest batch combines photometric saturation with physical liquid volume. To determine total yield, the Keeper records the average Candon reading per Dram of fluid drawn from the manifold and multiplies it by the total volume of fluid secured in the flasks. For example, if a collection run yields forty standard Lumium Flasks filled to their capacity of fifty Drams each, the total liquid volume is two thousand Drams. If the Candon-meter indicates an average density of ten Candons per Dram across the testing samples, the Keeper multiplies two thousand Drams by ten Candons per Dram to arrive at a verified harvest yield of twenty thousand Candons. This figure is entered into the master ledger and serves as the legal basis for guild taxation and trade credit.

4.4. Safety Protocols During Harvest

Working beneath a swirling canopy of bioluminescent insects carries significant physiological risks that demand strict protective measures. Guild regulations strictly prohibit any Keeper or apprentice from entering the harvest staging area without wearing the full mandatory protective gear. This equipment includes heavy steer-hide gauntlets to prevent skin irritation from condensed lumium, a full-length linen smock treated with fire-retardant salt, and specialized tinted goggles with a Candon-rating of at least two hundred Candons. Wearing goggles that fall below this two-hundred-Candon threshold fails to block the concentrated optical

rays bouncing off the prismatic ceiling panels during an active sweep.

The primary hazard confronting the harvest crew is an acute optical injury known as light-burn. Direct exposure to raw, un-flasked luminescence at close range causes light-burn, which temporarily blinds the keeper. The cold photons emitted by concentrated lumium possess a short, high-energy wavelength that rapidly bleaches the rhodopsin pigments inside human retinal rods. If a worker removes their protective eyewear while handling open collection pans or peering into the vacuum collar, the intense glare overstimulates the optic nerve, causing searing headache, phantom violet flashes, and sudden loss of central vision that can persist for up to forty-eight hours.

When a worker suffers accidental exposure, colleagues must execute immediate first aid to prevent permanent optical trauma. First, the afflicted individual must be guided out of the glass house immediately and into a completely blacked-out infirmary room. Second, the caregiver cleanses the victim's eyes by flushing them with sterile, chilled saline solution for five minutes to soothe irritated corneal tissues. Third, the caregiver applies a thick poultice of cold, crushed moon-root pulp wrapped in sterile gauze directly over both closed eyelids, securing the dressing with a dark linen bandage. The patient must remain in absolute darkness under this soothing herbal compress for twelve continuous hours, allowing retinal pigments to regenerate before their eyes are re-exposed to soft candlelight.

4.5. Processing and Transporting the Yield

Harvested luminescence cannot simply be loaded onto mountain pack wagons immediately following collection. The liquid must undergo three distinct processing phases before transport: settling, acoustic stabilization, and thermal grading. In the settling phase, filled flasks rest undisturbed in dark racks for six hours while heavy particulate matter drops to the bottom reservoir. In the second phase, acoustic stabilization, the flasks pass through a water bath subjected to steady chime tones, aligning the molecular clusters of the suspended lumium to reduce internal kinetic tension. In the final phase, thermal grading, inspectors verify that the fluid rests at an equilibrium temperature of forty Vels, assigning a quality grade based on its clarity and Candon-density.

Careful transport handling is critical because the refined fluid is exceptionally fragile. Stored luminescence degrades by two percent per hour if subjected to heavy physical vibration during transit. Excessive jostling and road shocks agitate the bound molecular rings of the suspended lumium, causing them to dissociate into non-luminescent components. A cart traversing rough mountain paths without adequate shock absorption can lose nearly half its light value before reaching Oakhaven. To prevent this financial loss, guild wagons are fitted with suspended carriage cradles lined with thick felt, and drivers are required to travel along paved valley turnpikes at a walking pace not exceeding four miles per hour.

Before any shipment departs the mountain facility, the supervising Luminary must draft a comprehensive transport manifest for the consignment. For a shipment of Grade A Lumium, the document must record five mandatory data points: the unique batch number, the total quantity of certified flasks, the verified Candon yield per flask, the departure timestamp, and the registered seal imprint of the inspecting Luminary. For example, a standard crate containing twenty Grade A Lumium Flasks of fifty Drams each, holding a tested value of fifteen Candons per Dram, will display an entry of twenty vessels containing one thousand Drams total, certified at fifteen thousand Candons. The transport manifest is signed in indelible ink, stamped with the

wax guild crest, and locked into a weatherproof tin cylinder handed directly to the convoy master.

5. Pests, Diseases, and Troubleshooting

Learning objectives

LO-5.1.1 Identify the physical markings of the Glass-Borer Beetle. (K1)

LO-5.1.2 Explain the life cycle of the Glass-Borer and how it breaches glass houses. (K2)

LO-5.1.3 Design a trap system to monitor for Glass-Borer incursions. (K3)

LO-5.2.1 List the first three visible symptoms of Wing-Fade Disease. (K1)

LO-5.2.2 Explain how nectar spoilage contributes to the spread of Wing-Fade. (K2)

LO-5.2.3 Implement quarantine protocols for a swarm showing Stage 2 Wing-Fade. (K3)

LO-5.3.1 Define "swarm panic" in the context of moth behavior. (K1)

LO-5.3.2 Explain how sudden temperature spikes trigger swarm panic. (K2)

LO-5.3.3 Execute the emergency dimming procedure to calm a panicked swarm. (K3)

LO-5.4.1 Recall the smell associated with fermented nectar spoilage. (K1)

LO-5.4.2 Discuss the chemical breakdown of moon-root in spoiled nectar. (K2)

LO-5.4.3 Neutralize a contaminated feeding trough using V-Scrub solution. (K3)

LO-5.5.1 State the maximum pressure limit before emergency venting is required. (K1)

LO-5.5.2 Explain the sequence of opening vents to prevent a vacuum collapse. (K2)

LO-5.5.3 Perform a manual override of a jammed apex vent. (K3)

Even the most carefully constructed glass house remains vulnerable to biological invaders, pathogenic outbreaks, and mechanical crises. A Keeper must act as both physician and guardian to the colony, spotting early warning signals before a localized disruption escalates into mass mortality. This chapter covers the detection and eradication of destructive predators, the containment of wing degeneration, the suppression of thermal panic, decontamination of tainted feeding troughs, and the manual venting procedures required during extreme barometric surges.

5.1. Identifying the Glass-Borer Beetle

The most dangerous external threat to any captive colony in the Alabaster heights is an invasive predator known as the Glass-Borer Beetle. Keepers must recognize the distinctive morphological features of this insect at a glance to prevent structural breaches. An adult beetle measures roughly two inches in length and features a heavily armored, oil-sheened carapace marked by alternating chevron stripes of charcoal and iridescent ochre. Its most telltale feature is a pair of serrated, steel-gray mandibles reinforced with bio-accumulated silicates, protruding well forward of its compound eyes. In addition, its hard wing cases display six raised, golden tubercles that rattle together when the beetle feels threatened, producing a low buzzing click audible across a quiet glass house.

The developmental cycle of this pest presents an extraordinary danger to the structural envelope of the

sanctuary. Adult beetles lay microscopic eggs within the tiny crevices where Copper-Resin muntins join the stone foundation or wooden pillars. Upon emergence, the larvae secrete an acidic enzyme capable of slowly dissolving resinous putties and soft sealants. As they mature into ravenous grubs, they chew directly through weakened seals between the window panes to gain entrance to the warm interior. Once inside, the adult beetles seek out resting clusters of moths, consuming their wing bases and biting through the thoracic lumium glands to gorge on concentrated alkaloids. Furthermore, if even a single Glass-Borer Beetle is detected inside the enclosure, Guild rules dictate that the entire moth colony must be placed on an iron-heavy diet for seven continuous days to deter the predator through unpalatable metallic hemolymph.

Monitoring for potential incursions requires an effective physical barrier and early-warning array. To establish a perimeter defense, the Keeper installs baited pitfall traps every ten feet along the exterior base of the Alabaster-Stone foundation wall. Each trap consists of a smooth-sided copper cylinder sunken level with the soil, containing twenty Drams of fermented sugar-sap spiked with mineral sea-ash, which mimics the scent of an open feeding trough. Above each pitfall, the Keeper mounts a sheet of clear glass coated with sticky pine pitch. Approaching beetles fly toward the aromatic lure, collide with the vertical pane, and tumble downward into the copper basin where the slick metal sides prevent their serrated legs from gaining traction. Checking these monitoring traps daily allows the Keeper to detect wild beetle populations moving down from the mountain slopes long before any masticated pane seals appear.

5.2. Wing-Fade Disease

5.2.1. Early Symptoms

When biological pathogens take hold within an enclosed colony, the deterioration is swift and visually unmistakable. The onset of Wing-Fade Disease is characterized by three initial visible indications that every apprentice must learn to identify. The first visible sign of the affliction is a chromatic alteration, wherein the insect's light shifts from its natural pale blue to a sickly, dull green. Within twelve hours of this color transformation, the second symptom emerges: the outer trailing edges of the moth's wings turn brittle, curling upward into dry, translucent fringes. The third early symptom follows shortly after, presenting as small, milky-white chalky spots that spread symmetrically across the thoracic chitin, signaling that internal tissue degeneration has begun.

5.2.2. Spoilage and Transmission

The underlying etiology of this disease is directly linked to substandard feeding hygiene within the facility. When synthetic nectar rests beyond its safe window or is brewed using contaminated vessels, virulent mold spores proliferate throughout the liquid. As nectar spoils, it forms toxic byproducts that erode the digestive lining of the moth's midgut. This intestinal damage compromises the insect's immune barriers, allowing opportunistic pathogens to enter the hemolymph and travel directly to the wing vein matrices and lumium glands. Because adult moths engage in communal feeding at gravity drip dispensers, infected individuals transfer these microscopic pathogens to the brass siphon tips through oral secretions. In this manner, an uncleaned dispenser can infect hundreds of healthy insects in a single night.

5.2.3. Quarantine Protocols

Containing an active outbreak demands rapid isolation to prevent total colony loss. The moment an outbreak reaches Stage Two, indicated by curling wing margins and spreading chalky spots across multiple insects, strict isolation measures must be enforced. However, an Apprentice or Journeyman cannot declare an isolation on their own authority; a Master Luminary is the only rank authorized to issue an official quarantine order. Once the order is signed, Keepers don protective gauntlets and use the silk Lunar-Net to capture infected moths, relying on the static-free weave to transport them into separate quarantine chambers without inflicting further mechanical trauma on their brittle wings. The entire infected population must remain in absolute isolation for a minimum of three complete cycles, which equates to exactly twenty-one days, during which they receive specialized medicated feed while the main glass house undergoes complete sanitation.

5.3. Managing Sudden Swarm Panic

Under normal conditions, a colony moves in synchronized harmony, but environmental disruption can shatter this order in seconds. In the context of moth husbandry, swarm panic is defined as an acute, collective hysteria wherein the insects lose their navigational schooling instincts, scattering into frantic, undirected flight at maximum speed. During an episode of panic, thousands of moths abandon their routine canopy patrol or feeding drift, colliding violently with structural Iron-Oak pillars, bouncing off the perimeter glazing, and crashing into one another. This blind, frenzied flapping leads to immediate wing breakage, massive scale shedding, and rapid physical exhaustion that can kill a quarter of the swarm within an hour if left unchecked.

The primary environmental catalyst for this dangerous behavioral breakdown is a sudden spike in ambient heat. While the colony thrives at forty-two Vels, the insects experience sudden swarm panic if the internal temperature reaches fifty-five Vels. At this critical thermal threshold, the insect's neural synapses fire erratically and the lumium oxidation inside their thoraxes accelerates out of control, flooding their sensory organs with intense internal glare. The moths perceive this runaway internal reaction as an immediate existential threat, triggering a desperate, blind escape response. In poorly managed enclosures, this catastrophic spike often happens on hot afternoons if south-facing Lunar-Tinted Glass has been improperly cleaned or if cooling vents fail during the dusk transition.

Halting an active panic event requires the Keeper to execute the emergency dimming sequence within minutes of the initial frenzy. First, the Keeper immediately pulls the external canvas blackout shades over the prismatic roof panels, cutting off all overhead light shafts. Second, the Keeper pulls the master damper lever on the floor to introduce a blast of cool subterranean air, dropping the chamber temperature below forty Vels. Third, the Keeper introduces a soothing, vaporized mist of lavender-infused spring water through the overhead misting nozzles at ninety Torr-Pascs of line pressure. Deprived of high light and cooled by the fine floral vapor, the moths' hyperactive flight muscles relax, their internal photogenic pulsing slows, and the insects safely settle onto the lower Iron-Oak branches in motionless resting clusters.

5.4. Dealing with Nectar Spoilage

The chemical stability of liquid feed is fragile, and offering degraded nourishment to a swarm invites immediate disaster. Keepers must conduct regular sensory audits of all storage casks and feeding stations. The most reliable early warning of degraded nectar is its distinctive olfactory signature: spoiled nectar produces an unmistakable, sharp odor resembling burnt almonds. This pungent aroma is accompanied by a thinning of the fluid's viscosity and the appearance of a foamy, gray scum along the liquid surface. The

burnt-almond scent is caused by airborne wild bacteria consuming simple sugars and generating toxic cyanide-derivative alcohols, signaling that the nectar has crossed the line from beneficial fermentation into lethal toxicity.

The underlying biochemical devastation caused by this spoilage centers on the destruction of the active tuber compounds. In spoiled nectar, the chemical breakdown of moon-root occurs as destructive bacterial enzymes cleave the delicate ring structures of the precursor alkaloids. Instead of binding smoothly with light alcohols into an organo-luminescent compound, the broken alkaloids degenerate into insoluble acidic crystals and corrosive gas. When ingested, these crystalline degradation products tear the delicate microvilli of the insect's digestive tract, completely blocking the absorption of iron and lumium. Starved of functional nutrients, the moth's light dims prematurely, and its body falls victim to opportunistic fungal spores.

Whenever a feeding station exhibits this burnt-almond odor, the Keeper must execute an immediate decontamination protocol using certified neutralizing agents. The Keeper removes the contaminated Sponge Trough or draining reservoir from its timber mount. Next, the Keeper scrubs every interior wooden and brass surface with pure, undiluted V-Scrub solution, using a stiff bristle brush to scour away bacterial films, mold residues, and hardened sugar crusts. However, because V-Scrub is an aggressive acidic cleaner, it must never be mixed with raw lumium, as doing so triggers a violent, highly exothermic reaction that can crack glassware and release caustic fumes. After scrubbing, the Keeper thoroughly flushes the trough with fifty Drams of pure boiling spring water, tests the rinse runoff with litmus paper to verify neutrality, and allows the wood to air-dry before inserting a fresh sea sponge and clean nectar.

5.5. Emergency Venting Procedures

During severe alpine thunderstorms, atmospheric pressure around a mountain glass house can fluctuate violently. While the manual Draft-Bellows handle small daily shifts, catastrophic pressure surges require rapid mechanical venting to preserve the building's structural integrity. Guild engineering standards dictate that emergency venting procedures must be initiated immediately if internal pressure exceeds one hundred and eighty Torr-Pascs of atmospheric resistance. Allowing the internal pressure to climb beyond this one-hundred-and-eighty-Torr-Pasc threshold puts immense outward force on the perimeter muntins, risking the explosive blowout of hundreds of square feet of delicate Lunar-Tinted Glass.

When venting becomes necessary, the order in which airflow hatches are opened is a matter of life and death for the architectural envelope. The Keeper must execute a precise venting sequence: the low north-wall louvers and mid-level baffle slots must always be opened completely before releasing the overhead exhaust hatches. Opening the apex vent before the floor vents during an emergency will cause a sudden vacuum collapse, shattering the Prismatic Panels into thousands of sharp quartz shards that would rain down and obliterate the swarm below. By opening the bottom louvers first, high-pressure air at ground level creates an outward thermal buffer, ensuring that when the apex is unlatched, pressure equalizes smoothly throughout the vertical column without creating destructive implosive vacuums.

If the high apex exhaust lever jams during an emergency surge due to ice accumulation or distorted Copper-Resin muntins, the Keeper must carry out an immediate manual override. The Keeper dons a safety harness and climbs the interior Iron-Oak service ladder to the high catwalk beneath the roof ridge. Armed with a brass spanner and an Iron-Oak wedge, the Keeper first disengages the mechanical linkage pin that

connects the stuck pull-rod to the central louver gear. Next, the Keeper inserts the wooden wedge into the seam between the central louver leaf and the frame. Striking the wedge firmly with a copper mallet breaks the ice seal and forces the vent leaf open by three inches, instantly releasing excess pressure into the storm. The Keeper then secures the lever in place using a safety chain, verifying on the catwalk barometer that chamber pressure drops below one hundred and forty Torr-Pascs before descending.

6. Keeper Ranks and Guild Regulations

Learning objectives

LO-6.1.1 List the four ranks within the Keepers' Guild. (K1)

LO-6.1.2 Describe the responsibilities of a Master Luminary. (K2)

LO-6.1.3 Determine the appropriate chain of command to report a structural failure. (K3)

LO-6.2.1 State the minimum number of harvest cycles required for certification. (K1)

LO-6.2.2 Explain the purpose of the Shadow-Trial during apprenticeship. (K2)

LO-6.2.3 Prepare a portfolio of brewing logs for a Journeyman review. (K3)

LO-6.3.1 Identify the three primary tools issued to an Apprentice. (K1)

LO-6.3.2 Explain the multi-purpose function of the Lunar-Net in harvesting and pest control. (K2)

LO-6.3.3 Maintain and recalibrate a standard-issue Candon-meter. (K3)

LO-6.4.1 List the five key areas inspected during a Guild Audit. (K1)

LO-6.4.2 Discuss the penalties for unauthorized glass tinting. (K2)

LO-6.4.3 Conduct a pre-audit self-assessment of the brewing station. (K3)

LO-6.5.1 Recite the first tenet of the Keeper's Code. (K1)

LO-6.5.2 Explain the ethical implications of over-harvesting a dormant swarm. (K2)

LO-6.5.3 Evaluate a hypothetical dispute between two Keepers over vent sharing. (K3)

The care of Lepidoptera luminis is not merely an agricultural pursuit, but a sacred trust overseen by an ancient fraternity. From its grand hall in Oakhaven, the Keepers Guild establishes the laws, operational qualifications, craft tools, and ethical tenets that govern every glass house in the Alabaster range. This final chapter outlines the ladder of promotion, the rigorous examination of initiates, the care of issued equipment, the statutory audit of facilities, and the moral covenant that places the welfare of the swarm above all commercial gain.

6.1. The Hierarchy of the Keepers' Guild

The governing structure that oversees the husbandry of light is organized into four distinct grades of progressive mastery. Arranged from lowest to highest, these grades are Apprentice, Journeyman, Luminary, and Master Luminary. Every individual entering the service of the moths starts as an Apprentice, performing the physical labor of cleaning troughs, recording temperatures, and carrying casks under constant instruction. Upon passing practical trials, an initiate rises to the rank of Journeyman, earning the right to supervise independent glass houses and manage daily feeding schedules. Higher still stands the Luminary, an expert craftsman who directs harvest sweeps, certifies transport manifests, and trains new intakes. At the apex sits the council of Master Luminaries, who guide Guild policy, oversee regional distribution, and maintain statutory discipline throughout the mountains.

The authority vested in a Master Luminary carries immense operational and legal weight across the

provinces. Rather than handling mundane physical tasks, a Master Luminary holds exclusive responsibility for high-level regulatory decisions that affect regional colony health. For example, a Master Luminary is the only rank authorized to issue an official quarantine order when Wing-Fade Disease strikes an enclosure, legally sealing the perimeter against all external contact. Furthermore, Master Luminaries govern architectural certifications, approve major glass house construction blueprints, preside over formal disciplinary tribunals, and administer the final sacred oaths to advancing craftsmen. Without the explicit written wax seal of a Master Luminary, no facility may expand its footprint, import breeding stock from wild mountain valleys, or market Grade A Lumium in Oakhaven.

Navigating this institutional structure during an emergency requires absolute obedience to established protocol. If an on-duty craftsman discovers a critical structural failure, such as a cracked load-bearing beam or buckling wall footing, they must adhere to a strict reporting sequence. An Apprentice who spots sagging Copper-Resin ribs or a fractured Iron-Oak pillar must immediately notify their supervising Journeyman. The Journeyman logs the coordinates of the damage, secures the immediate area, and escalates the report directly to the resident Luminary. If the Luminary verifies that the timber compromise threatens canopy stability or risks a catastrophic pressure rupture, they must bypass local operations and dispatch an emergency courier to the presiding Master Luminary in Oakhaven. Attempting to repair a primary load-bearing pillar without notifying this chain of command constitutes a grave regulatory breach.

6.2. Apprenticeship and Certification

Ascending the guild ladder from raw recruit to verified Journeyman requires dedication, physical endurance, and verifiable documentation of skill. An Apprentice cannot simply request promotion based on time served; advancement is tied directly to practical field experience. According to Guild statutes, an Apprentice must complete exactly twelve harvest cycles before they are eligible to take the Shadow-Trial. Each cycle corresponds to a full four-week generation of the captive swarm, meaning an initiate must spend nearly a full year actively working catwalk sweeps, calibrating meters, and handling vacuum flasks before their petition for advancement is considered by the examining board.

The gateway assessment for every aspiring craftsman is the legendary examination known as the Shadow-Trial. The purpose of this rigorous evaluation is to test the initiate's sensory acuity, diagnostic instincts, and composure when deprived of sight. In the darkened testing vaults beneath Oakhaven, the candidate is blindfolded and presented with three unmarked, resting nectar vats at varying stages of fermentation. To succeed, the candidate must identify three different nectar spoilage stages by smell alone. In particular, the candidate must accurately distinguish between the rich, yeasty bouquet of properly aged feed, the flat sweetness of unfermented sap, and the sharp, dangerous scent of burnt almonds that signals lethal bacterial decomposition. An Apprentice who misidentifies the almond aroma of spoiled nectar fails immediately, as distributing such feed would poison hundreds of moths under their care.

Before appearing before the council for the oral examination, the applicant must compile a meticulous technical portfolio documenting their culinary precision. To prepare a portfolio of brewing logs for a Journeyman review, the candidate must assemble fifty consecutive batches of handwritten production records. Each individual log sheet must document seven mandatory parameters: the date and hour of initial soaking, the exact weight of crushed moon-root, the volume of spring water used, the simmer temperature

recorded in Vels, the ambient relative humidity at the time of cooking, the fermentation resting duration in hours, and the final specific gravity of the filtered liquid. The candidate must bind these fifty verified entries between wooden covers crafted from Iron-Oak bark, stamped by their supervising Luminary, demonstrating consistent recipe replication without a single spoiled batch.

6.3. Tools of the Trade

Upon formal induction into the craft, every Apprentice is entrusted with a standardized kit of mechanical implements designed for daily glass house operations. Guild quartermasters issue three primary tools upon induction: a Candon-meter, a Lunar-Net, and a pair of Draft-Bellows. These three devices represent the essential sensory, harvesting, and atmospheric controls required to manage a colony. The Apprentice is personally accountable for the preservation of these instruments; losing an issued tool or allowing its brass fittings to corrode results in immediate disciplinary labor in the stone foundations.

While primarily intended for collecting ethereal radiance under the high roof, the Lunar-Net serves an essential secondary purpose in maintaining biological safety within the glass house. Its frame consists of a polished Iron-Oak hoop holding a delicate mesh woven from static-resistant silk and fine silver filaments. During the Luminous Harvest, its soft wake and anti-static properties gather airborne light vapors without injuring flying adults. In pest control and disease containment, however, this same net becomes an indispensable medical tool. When a colony suffers an outbreak of Wing-Fade Disease, the Keeper must use the soft Lunar-Net to capture infected moths safely without tearing their brittle, curling wings. Furthermore, the silver-threaded weave repels wild fungal spores and prevents cross-contamination between quarantine chambers, allowing workers to isolate diseased specimens smoothly.

To ensure measurements remain legally and scientifically valid, every Keeper must perform daily calibration and maintenance on their issued Candon-meter. The procedure begins each morning in the tool room by cleaning the quartz diffusion lens with a soft flannel cloth to remove all dust and nectar mist. Next, the Keeper places the meter's selenium sensor plate inside an insulated test chamber filled with pure synthetic nectar that has been chilled until slush crystals appear on its surface, indicating the exact freezing point of twelve Vels. Because pure nectar emits zero luminous flux at this baseline temperature, the meter's analog pointer must align perfectly with the zero-Candon hash mark. If the needle deviates above or below zero, the Keeper inserts a tiny brass jeweler's screwdriver into the adjustment port behind the dial, turning the balance spring screw until the indicator rests true, before locking the retention ring.

6.4. Guild Inspections and Audits

6.4.1. Preparing for the Audit

To maintain universal standards of safety and animal welfare, Guild inspectors conduct unannounced statutory audits of every licensed glass house twice each calendar year. When preparing for an official audit, a Keeper must ensure five key operational areas are in immaculate order: structural frame integrity, ventilation baffle mechanics, brewing station sanitation, medicine and quarantine readiness, and the accuracy of harvest ledgers. Inspectors verify that all Iron-Oak pillars remain free of rot and beetle borings, check that louver cables operate freely without binding, examine feeding troughs for traces of bacterial residue, inspect emergency supplies of V-Scrub and moon-root poultices, and audit ledger entries against stored Lumium

Flask serial numbers.

6.4.2. Common Infractions

During these regulatory reviews, inspectors cite facilities for various maintenance failures, but few violations are treated with greater severity than unauthorized glass modification. The architectural integrity of south-facing walls depends on genuine Lunar-Tinted Glass containing suspended metallic silver to scatter harsh solar rays. If a Keeper attempts to cut costs by installing clear soda-lime glass, or applies unauthorized chemical surface tints, varnishes, or stained glazes without Guild certification, the penalties are catastrophic. Unauthorized glass modification results in immediate suspension of the facility's operating charter, confiscation of the current harvest yield, and the mandatory demotion of the presiding Keeper down to the rank of Apprentice. Such strict measures prevent unscrupulous managers from blinding swarms with cheap, unfiltered daylight.

6.4.3. Pre-Audit Self-Assessment

To avoid citations and disciplinary penalties, a prudent Keeper conducts a rigorous pre-audit self-assessment of the brewing station twenty-four hours before the scheduled inspection window. The Keeper begins by draining and inspecting the copper simmering kettles, checking the interior walls for greenish verdigris or scorched mineral scale. Next, the Keeper scrubs all unglazed earthenware soaking jars with pure boiling spring water, verifying that no residual moon-root pulp remains lodged in the ceramic pores. The Keeper then examines the oak fermentation casks, inspecting each wooden bung and spigot for mold or weeping syrup, and discards any batch exhibiting an abnormal aroma. Finally, the Keeper verifies that the brass balance scales are zeroed, calibration weights are polished, and the current brewing log book is signed and dated through the morning shift, ensuring inspectors find an orderly, compliant facility.

6.5. The Keeper's Code of Ethics

Above all technical skills, architectural codes, and commercial quotas stands the moral framework of the craft, embodied in the sacred Keeper's Code of Ethics. Established during the founding of Oakhaven, this charter reminds every craftsman of their sacred duty toward the living creatures under their care. Every initiate must memorize and internalize the foundational core of the doctrine. The first tenet of the Keeper's Code states: "The light serves the house, but the Keeper serves the moth." This sentence defines the fundamental hierarchy of values within the craft: while the radiant energy produced by the swarm brings warmth and illumination to human settlements, the physical wellbeing and dignity of the insects must never be sacrificed for financial gain or human convenience.

This ethical principle faces its greatest test during the harsh winter months of the tenth calendar cycle. When the Frost Moon rises over the Alabaster Mountains, the captive insects instinctually enter a thirty-day metabolic suspension, ceasing all egg laying and resting silently along the Iron-Oak beams. Stripping the canopy of luminescence during this natural rest period constitutes a severe moral and biological crime. Harvesting light from a dormant swarm during the Frost Moon directly violates Section Four of the Keeper's Code of Ethics. Because dormant moths operate on severely depleted metabolic reserves, forcing them into flight with harvest nets depletes their remaining hemolymph sugars, causing catastrophic spring mortality. The Guild views this practice as unconscionable greed; any craftsman caught harvesting during the sacred

month is permanently stripped of their rank, expelled from the Guild, and banished from the mountain valleys.

When practical conflicts arise between neighboring facilities, ethical principles provide the framework for dispute resolution. Consider a scenario where two adjacent glass houses share a common central ventilation flume between their Iron-Oak walls. During an unexpected summer heat wave, the Keeper of House North attempts to close the shared exhaust damper to force cooler updrafts through their own lower louvers, which consequently traps fifty-five-Vel heat inside House South and drives their swarm into a violent panic. When evaluated under Guild ethical standards, the action of Keeper North is judged as reckless endangerment of living stock. Guild statutes mandate that in all shared-utility systems, common dampers must remain locked at a neutral fifty-percent aperture during climate extremes. A Keeper must endure minor yield reductions in their own facility rather than take unilateral actions that risk thermal panic or structural damage in a neighboring sanctuary.